SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Actuarial Mathematics and Statistics

F78PA
Probability and Statistics A

Friday 14 December 2012

1.30 pm to 3.30 pm

Question	Marks
1	11
2	10
3	17
4	7
5	14
6	14
7	12
Total Marks	85

Candidates must attempt ALL SEVEN questions
and show their working.

Approved electronic calculators may be used

AMS Formula Sheet and Cambridge Statistical Tables are provided

INSTRUCTIONS: Attempt ALL SEVEN questions. To receive full credit you must show your work and explain your answers.

1. Sketch the functions given below and determine which of the functions are density functions. Give reasons for your answers, and, for each *density*, find the corresponding *distribution function*.

(a)

$$f(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-x}, & 0 \leq x. \end{cases}$$

(b)

$$g(x) = \begin{cases} 3(x-1)^2, & 1 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

(c)

$$h(x) = \begin{cases} \frac{1}{4}, & 0 < x < 1 \\ \frac{3}{4}, & 1 \leq x < 2 \\ 0, & \text{otherwise.} \end{cases}$$

[Total 11 marks]

2. Suppose that X is a continuous random variable with density function given by

$$f_X(x) = \begin{cases} Ax + B & -1 \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

where A and B are constants. Also, suppose that $E(X) = \frac{1}{3}$

(a) Use the information above to show that $A = B = \frac{1}{2}$. [2 marks]

(b) Using the density function $f_X(x)$, or otherwise, determine the conditional probability $\mathbf{P}\{X \leq 1/2 \mid X \geq 0\}$. [3 marks]

(c) Suppose that $Y = |X|$. Show that $Y \sim U[0,1]$ (i.e show that Y is uniformly distributed on the interval $[0,1]$).

[5 marks]

[Total 10 marks]

[PLEASE TURN OVER]

3. A point (X, Y) is picked **uniformly** in a triangle with corners given by the points (0,0), (4,0) and (0,2). The joint density of X and Y is given by

$$f(x,y) = \begin{cases} \frac{1}{4} & x, y \geq 0,\ x + 2y \leq 4, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Sketch the region where the joint density $f(x, y)$ does not equal zero. [1 mark]

(b) Determine the marginal density $f_Y(y)$ for Y, and hence, or otherwise, determine $\mathbf{P}(Y > 1)$.

[4 marks]

(c) Using a geometric argument, or otherwise, determine the conditional probability $\mathbf{P}(X > 1 \mid Y > 1)$.

[5 marks]

(d) Let $Z = X + 2Y$. By considering $\mathbf{P}(X + 2Y \leq z)$, determine $F_Z(z)$, the distribution function of Z. Hence, or otherwise, determine $f_Z(z)$, the density function for Z. [7 marks]

[Total 17 marks]

4. Suppose that X is a random variable with (marginal) density

$$f_X(x) = \begin{cases} 2x & 0 < x < 1, \\ 0 & \text{otherwise.} \end{cases}$$

and that Y is a random variable with conditional density, given $X = x$,

$$f_{Y|X}(y|x) = \begin{cases} xe^{-xy} & 0 < x < 1, 0 < y < \infty, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Determine the joint density $f_{X,Y}(x, y)$ for the variables X and Y and hence, or otherwise, determine $\mathbf{P}(Y > X)$. [5 marks]

(b) **Without calculation**, explain why $E(Y \mid X = x) = \frac{1}{x}$ when $0 < x < 1$. [2 marks]

[Total 7 marks]

[PLEASE TURN OVER]

5. Suppose that $X_1, X_2, \ldots, X_n$ are *independent* random variables with distribution functions $F_1(x), F_2(x), \ldots, F_n(x)$ respectively. Let $M = \max\{X_1, X_2, \ldots, X_n\}$.

(a) Let $F_M(x)$ denote the distribution function for M. Show that for any $x \in \mathbb{R}$,

$$F_M(x) = F_1(x) \cdot F_2(x) \cdot \cdots \cdot F_n(x).$$

[4 marks]

(b) Sam has two alarm clocks which go off each morning, *independently of each other*, at random times which are uniformly distributed between 7 and 8 am. Let X be the time, measured as minutes past 7 am, at which one of the clocks rings (so $X \sim U[0, 60]$) and let Y be the time, measured as minutes past 7 am, at which the other clock rings (so $Y \sim U[0, 60]$).

Suppose that Sam gets up as soon as both clocks have rung and let $T = \max\{X, Y\}$ denote the time, in minutes past 7am, that Sam gets up.

i) Determine F_T, the distribution function of T. [3 marks]

ii) Determine the *median* of the distribution function F_T. [2 marks]

iii) To catch the bus to university, Sam must leave his flat by 8 am. If it takes Sam 10 minutes to get ready to leave, find the probability that he misses the bus. [2 marks]

iv) Determine the *conditional* probability that Sam misses the bus, *given* that Sam has not got up by 7.30 am. [3 marks]

[Total 14 marks]

6. (a) Suppose that $X_1, X_2, \ldots$ are i.i.d. random variables with common moment generating function $M_X(\theta)$ and suppose that N is a counting variable with probability generating function $G_N(s)$. In addition, suppose that N is independent of $X_1, X_2, \ldots$ and let $S_N = X_1 + X_2 + \cdots + X_N$. Show that the moment generating function of S_N is given by

$$M_{S_N}(\theta) = G_N(M_X(\theta)).$$

Hence, using a moment generating function argument, show that $E(S_N) = E(N)E(X)$. [7 marks]

(b) Suppose that the number, N, of accidents on a building site in one year has a Poisson distribution with mean equal to 15 accidents per year. An accident is categorised as 'serious' if those involved in the accident must be sent to the hospital. Past experience has shown that the probability that an accident is 'serious' is $p = 0.12$, independent of all other accidents and independent of the number of accidents, N, in one year.

Use a moment generating argument to determine the distribution of T, the number of 'serious' accidents is one year and hence, compute $\Pr\{T > 1\}$. [7 marks]

[Total 14 marks]

7. A clerk in a small insurance company returns from holiday to discover that he has a stack of 50 claims to process. Based on past experience, the clerk estimates that the time to process each claim (in minutes) has a $\Gamma(5, 2)$ distribution, independent of the process times of all other claims.

Let X_1 denote the time that it takes to process the first claim in the stack, let X_2 denote the time that it takes to process the second claim in the stack, etc. and, in general, let X_n denote the time that it takes to process the nth claim.

(a) State carefully the Central Limit Theorem and use it to approximate the probability that, starting at 9am, the clerk manages to process all of the claims before 11am. [6 marks]

(b) Suppose that, independent of all other claims, the probability, p, that a claim exceeds £1000 is 0.3. Let Y equal the number of claims, out of the 50 that the clerk must process, that are greater than £1000.

State the exact distribution of Y and use the Central Limit Theorem to approximate $\Pr\{Y > 20\}$. [6 marks]

[Total 12 marks]

[END OF PAPER]